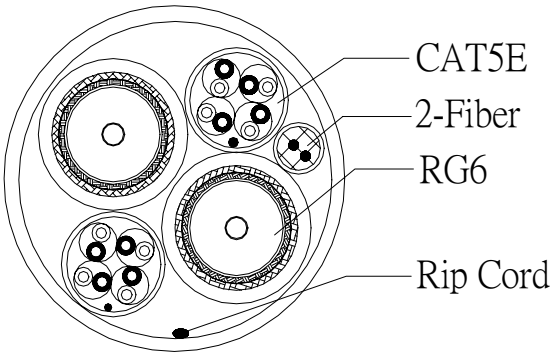
WAVENET HOME NETWORKING CABLE

Cross Section



P/N	Outer Jacket Color	Packing
70306	BLUE	500FT/REEL

Outer Jacket Marking

E223070 (UL) TYPE CM-OF 2 x RG6 QUAD 18AWG + 2 x CAT5E 350 & 62.5 UM MULTIMODE FIBER 75C --- ****** FEET mmyyd jnn

Description

Rated Temperature (^{o}C)	75
Product Standard Certification	UL
Flammability Test	CM

Two 4 Pair CAT5e UTP Cables
Complies to TIA/EIA 568B
24AWG Solid Bare Copper Conductor / PE Insulation
Two RG6 Quad Shield Coaxial Cables
Complies to UL1655, UL13, UL444
18AWG Bare Copper Conductors
Al foil / 60% Al Braid Shield + Al foil / 40% Al Braid Shield
One 2-Fiber Interconnect Cable
FDDI-Grade Fiber 62.5/125 um
Application
Multi-media Cables for Smart Home
Reference Standard
UL1655, UL13, UL444, TIA/EIA-568-B

Construction

2xCAT5E UTP LAN Cable
Detail See Attachment 1
2xRG6 Quad Shield Coaxial Cable
Detail See Attachment 2
2-Fible interconnect Cable
Detail See Attachment 3

Cabling Lay Length(±100mm)	380

PET Tape Wrapping

Outer Jacket	**PVC**
Average Thickness(±0.05mm)	0.85
Min. Point Thickness(mm)	0.76
Outer Dia.(±1.5mm)	17.5
Rip Cord	YES
Weight(kg/km)	226.4

Revision History — Rev. 0

Performance

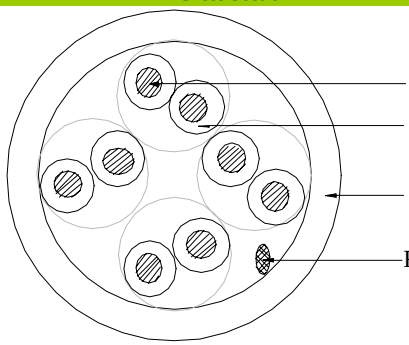
The Characteristics of RG6, CAT5e and Fiber See Attachment

Outer Jacket Mechanical Characteristics:

Test Object		
Test Material		PVC
Before Aging	Tensile Strength (Mpa)	>=13.8
	Elongation (%)	>=100
Aging Condition (^{o}C×hrs)		100 × 240
After Aging	Tensile Strength (Mpa)	>= 85% unaged
	Elongation (%)	>= 50% unaged
Cold Bend (-20±2°C×4hrs)		No crack

WAVENET HOME NETWORKING CABLE Attachment 1 ---- 4 Pair CAT5e UTP Cables

Cross Section



Marking

E223070 24AWG 4P 75C UTP --- (UL) CMR or C(UL) VERIFIED CAT.5E TO EIA/TIA 568-B.2 350MHz ****** FEET

Description

Reference Standard

UL Subject 444,EIA/TIA568-A & ISO/IEC 11801

Flammability Test	**CMR**

Construction

Conductor	**Solid Bare Copper**
AWG	24
Conductor Dia. (+0.008/-0.01mm)	0.525
Insulation	**PE**
Average Thickness(±0.03mm)	0.21
Min. Point Thickness(mm)	0.17
Insulation Dia.(±0.03mm)	0.93
Twisting Lay Length(mm)	30underneath
Cabling Lay Length(±20mm)	100
Inner Jacket	**PVC**
Average Thickness(±0.05mm)	0.50
Min. Point Thickness(mm)	0.43
Cable Dia.(±0.2mm)	5.00
Rip Cord	YES

Color

Insulation colors are:

Blue,White/Blue
Orange,White/Orange
Green,White/Green
Brown,White/Brown

Inner Jacket colors:

BLUE, YELLOW

Performance

Electrical Characteristics:

Frequency (MHz)	Return loss (dB)	Attenuation (dB/100m)	NEXT (dB)	ACR (dB)
0.772	19.4	1.8	67.0	65
1	20.0	2.0	65.3	63
4	23.0	4.1	56.3	52
8	24.5	5.8	51.8	46
10	25.0	6.5	50.3	44
16	25.0	8.2	47.3	39
20	25.0	9.3	45.8	37
25	24.3	10.4	44.3	34
31.25	23.6	11.7	42.9	31
62.5	21.5	17.0	38.4	21
100	20.1	22.0	35.3	13

Frequency (MHz)	PSNEXT (dB)	ELFEXT (dB/100m)	PSELFEXT (dB/100m)	Prop'n Delay (ns/100m)
0.772	64.0	66.0	63.0	575.0
1	62.3	63.8	60.8	570.0
4	53.3	51.7	48.7	552.0
8	48.8	45.7	42.7	546.7
10	47.3	43.8	40.8	545.4
16	44.3	39.7	36.7	543.0
20	42.8	37.7	34.7	542.0
25	41.3	35.8	32.8	541.2
31.25	39.9	33.9	30.9	540.4
62.5	35.4	27.8	24.8	538.6
100	32.3	23.8	20.8	537.6

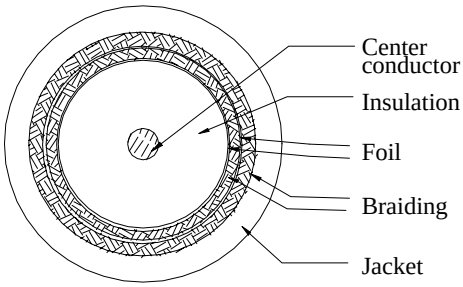
1.0-100.0MHz mpedance (ohms)	100 ± 15
1.0-100.0MHz Delay Skew (ns/100m)	>=45
Pair-to-Ground Capacitance Unbalance (pF/100m)	>=330
Max. Conductor DC Resistance 20°C (ohms/km)	93.8
Resistance Unbalance (%)	>=5

Mechanical Characteristics:

Test Object		**Inner Jacket**
Test Material		PVC
Before Aging	Tensile Strength (Mpa)	>=13.8
	Elongation (%)	>=100
Aging Condition (°Cxhrs)		100x240
After Aging	Tensile Strength (Mpa)	>=85% of unaged
	Elongation (%)	>=50% of unaged
Cold Bend(-20±2°Cx4hrs)		No crack

WAVENET HOME NETWORKING CABLE Attachment 2 ---- RG6 Quad Shield Coaxial Cables

Cross Section



Marking

E223070 RG-6/U QUAD SHIELD 18AWG 75C (UL) TYPE CM OR CL2
SWEEP TESTED TO 3.0GHZ ****** FEET

Description

Reference Standard
UL1655, UL13, UL444

Flammability Test	CM

Construction

Center Conductor	**Bare Copper**
AWG	18
Dia.	1.02
Insulation	**Skin Foam PE**
Nom. Thickness(mm)	1.78
Insulation Dia. (±0.08mm)	4.57
The first Al-maylar Shield (Overlapping, %)	>=25
The first Braid Shield	**Aluminium Wire**
Construction (mm)	16/4/0.16
Coverage Area (%)	>=60
The Second Al-maylar Shield (Overlapping, %)	>=25
The Second Braid Shield	**Aluminium Wire**
Construction (mm)	16/3/0.16
Coverage Area (%)	>=40
Inner Jacket	**PVC**
Nom. Thickness (mm)	0.75
Min. Thickness(mm)	0.62
Cable Dia.(±0.15mm)	7.50

Color

Inner Jacket Color:
White,Black

Performance

Electrical Characteristics:

Frequency (MHz)	Attenuation (dB/100m)
1	0.85
10	2.66
50	4.79
100	6.72
200	9.28
400	13.28
700	18.36
900	20.43
1000	21.61
1200	24.59
1450	26.36
1800	28.98
2200	32.03

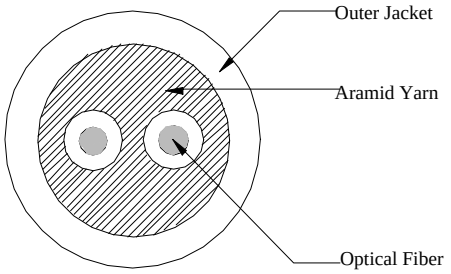
Dielectric Strength (kV/min)	1.0
Impedence (±3.0ohms)	75.0
SRL (dB,5~2200MHz)	>=20
Capacitance (pF/m)	53.1
Conductor DCR@ 20oC (ohms/km)	<=21.4
DC Loop Resistance@ 20°C (ohms/km)	<=136
Nom. Velocity Of Propagation (%)	82

Mechanical Characteristics:

Test Object		**Inner Jacket**
Test Material		PVC
Before Aging	Tensile Strength (Mpa)	>=13.8
	Elongation (%)	>=100
Aging Condition (°C×hrs)		100 × 240
After Aging	Tensile Strength (Mpa)	>= 85% unaged
	Elongation (%)	>= 50% unaged
Cold Bend (-20±2°C×4hrs)		No crack
Inner Jacket impact test(-15°C)		No crack
Inner Jacket Longitudinal Shrinkage (%)		<=5
Center Conductor Bond To dielectric (N)		>=2.3

WAVENET HOME NETWORKING CABLE Attachment 3 ---- 2-Fiber Interconnect Cable

Cross Section



Marking

E223070 (UL) OFNR DUPLEX 62.5/125 75C --- C(UL) OFNR DUPLEX 62.5/125 75C MULTIMODE OPTICAL CABLE (MM/YY) ****** FEET

Description

General Information

PVC tight buffered optical fiber, Riser type PVC Jacket

Meets NEC riser (OFNR) safety standards

Application

Intrabuilding backbones & horizontal installations in riser and general purpose environments

Product Standard Certification	UL
Reference Standard	UL1651, UL1666, TIA/EIA-568-B.3&Bellcore GR-409

Construction

Geometrical Characteristics:

2 Cores	Ultra Pure Silica Glass
Core Dia. (±3μm)	62.5
Clad Dia. (±2μm)	125
Cladding Non-Circularity (%)	<2.0
Core/Clad Concentricity Error (μm)	<=3.0
Primary Coating Material	**UV Curable Acrylate**
Primary Coating Dia. (±5μm)	245
Secondary Buffer Material	**PVC**
Secondary Buffer Dia. (±0.05mm)	0.9
Strength Member	**Aramid Yarn**
Jacket	**PVC**
Nom. Thickness (mm)	0.7
Min. Thickness (mm)	0.53
Cable Dia. (±0.15mm)	3.8
Weight (kg/km)	13.03

Color

Buffer Identification:

Blue,White

Jacket:

Orange

Performance

Optical Characteristics:

Numerical Aperture (±0.015)		0.275
Wavelength (nm)	At 850nm	At 1300nm
Max. Attenuation (dB/km)	<=3.5	<=1.5
Min. Modal Bandwidth (MHz.km)	>=200	>=500

Mechanical Characteristics:

Low and High Temperature Cable Bend		Mandrel Size of 50mm
Max. Tensile Load (N)	Short Term	700
	Long Term	450
Operation Temperature (^{o}C)		-20~75
Jacket Tensile Strength (Mpa)		13.8
Aging Conditions		110±2°C x 168h
Elongation-at-Break	Before Aging	>=166
	After Aging	>=100
Cable Aging		85±2°C x 168h
Min. Bending Radius (mm)	Short Term	45
	Long Term	50